Xiaochuan Li

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Research focus

I study language and cognition, especially how the structure of language shapes learning and thought. With language models as learners, which receive only controlled input and whose internals can be changed, these questions can be tested experimentally.

Education

Oct. 2025 – exp. 2027	**University of Technology Nuremberg**, Germany *M.Sc. Human and Artificial Intelligence* Coursework: deep learning, interpretability of language models, psychology and empirical study design, cognitive neuroscience.
Oct. 2024 – Jul. 2025	**University of Marburg**, Germany *German for graduate study; DSH-2 (C1 level)*
Sep. 2020 – Jun. 2024	**Shenzhen University**, China *B.A. Philosophy* Coursework: formal and modal logic, philosophy of language.

Research experience

Oct. 2026 – present	**The direction of disagreement: what departures from the majority term teach a text-only learner** *Independent project* Turned vagueness into a learning experiment: models read only naming sentences, with speakers' disagreement kept or randomly replaced, and are tested on whether they recover the order of categories. *Methods:* formal proof, transformers and LSTMs trained from scratch, fine-tuning, World Color Survey.
Sep. 2026 – present	**Modal language development in preschoolers** *M.Sc. thesis, Max Planck Institute for Human Cognitive and Brain Sciences* Behavioural study of how preschoolers acquire modal words such as *might* and *must*. In design.
Apr. – Nov. 2026	**Moral framing and dilemma choices in language models** *Project with the German Research Center for Artificial Intelligence (DFKI)* Led the utilitarian-framing branch: turned moral decisions into minimal pairs to measure how framing shifts language-model choices. Manuscript in preparation. *Methods:* probing, activation patching, low-rank analysis, steering.
Feb. 2026	**How the first information source shapes verification: a between-subjects experiment** *Small-scale psychology experiment* *Methods:* between-subjects design, Pavlovia (49 participants, 47 analysed), JASP.

May 2024	**Leibniz's logical system and characteristica universalis** *B.A. thesis, Shenzhen University* Leibniz's logic and his proposal for a universal symbolic language in which reasoning becomes calculation.

Preprints

Li, X. (2026). *The direction of disagreement: What departures from the majority term teach a text-only learner* [Preprint]. LingBuzz, lingbuzz/010504. https://lingbuzz.net/lingbuzz/010504

Research infrastructure

StemmaLM
Long-term infrastructure for language-model lineages

Turned questions about language development and transmission into model lineages raised on child-scale corpora and controlled variants: 60+ models, up to 98M parameters, public registry. *Methods:* training from scratch, minimal-pair evaluation, Slurm on an HPC GPU cluster. stemmalm.github.io

Employment

Apr. 2026 – present	**University of Technology Nuremberg**, Germany *IT Service Desk (part-time)* Student IT support; basic HPC user administration (resource allocation, access rights).
Sep. – Dec. 2021	**Shenzhen Learning Disorders Association**, China *Internship · Learning assessment at Yucai No. 2 Primary School* Ran reading assessments with primary-school children referred for learning difficulties.

Skills and languages

Human data	Experimental design, PsychoPy/Pavlovia, JASP
Evaluation	Controlled behavioural evaluation, minimal pairs
Training	Small transformers and LSTMs trained from scratch; fine-tuning
Interpretability	Activation patching, low-rank analysis, probing, steering
Tools	Python (PyTorch, Hugging Face Transformers, pandas, scikit-learn), Git, LaTeX, Slurm on an HPC GPU cluster
Languages	Mandarin Chinese (native), German (C1), English (C1), Japanese (learning)